DraftBench

An offline-first, air-gapped special education assistant

for primary school teachers in Nepal

100% local | No cloud | No telemetry | Nepali and English

WHY THIS MATTERS

The problem

On the ground

- Special educators are scarce in Nepali primary schools.
- General classroom teachers support learners with autism and other additional needs, often alone.
- Many schools have low or no connectivity.

What gets in the way

- Writing an IEP, adapting a lesson and preparing accessible materials takes hours teachers do not have.
- Cloud AI tools do not work offline.
- Cloud tools are unsuitable for sensitive student data.

FROM OBSERVATION TO MATERIALS

The solution

Teacher writes (Nepali or English):

"Aarav gets upset when the routine changes and can't hold a pencil for long."

IEP goals

Measurable, time-bound, tied to the observation

Autism-friendly lessons

Predictable, visual, low sensory load

Braille

Worksheets and lessons transcribed

NSL diagrams

Directional vectors for sign movement

Worksheets and quizzes

Aligned with CDC standards

Runs entirely on a school laptop. Minutes instead of hours, with student data never leaving the school.

CAPABILITIES

Key features

Observation to IEP

Free-form notes become structured, measurable IEP goals via an iterative planner loop.

Nepali and English

Language auto-detected and carried through every stage; Devanagari curriculum pages supported.

ASD rubric evaluation

Every lesson scored by a rule-based rubric; drafts below 8 are revised (up to 3 attempts).

CDC-aligned RAG

Local ChromaDB index with nomic-embed-text embeddings of CDC guidelines.

Cultural localization

Nepali currency, flora and fauna, festivals and regional contexts replace generic examples.

Local vector assets

Model-drawn SVG diagrams and NSL hand-sign schematics rendered by CairoSVG. No diffusion models.

Multi-format export

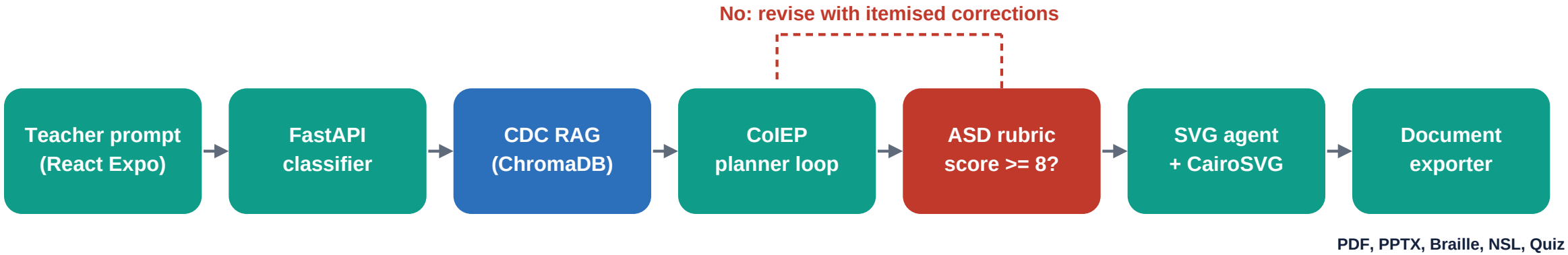
PDF, PPTX, Braille, NSL vectors and quiz sheets.

Cross-platform and secure

One Expo codebase for phones, tablets and web; local JWT auth with no external provider.

REQUEST PIPELINE

System architecture



Design decisions

- Offline-first by design: every inference, search, render and export runs locally.
- RAG over fine-tuning: curriculum updates mean re-ingesting documents, not retraining.
- Generate, evaluate, revise: a rubric gate compensates for the limits of a 7B model.

ORCHESTRATED BY UNIFIED_PIPELINE.PY

Agent pipeline in detail

1. **Teacher prompt**
   Informal observation plus optional grade, subject, profile and desired outputs.

2. **Intent classifier**
   Returns JSON: intent, topic, grade, subject, language. Routes the request.

3. **CDC RAG**
   query_curriculum_rag() retrieves grade and subject filtered curriculum chunks.

4. **CoIEP planner**
   Drafts lesson slides, then teacher scripts, accommodations and the IEP goal.

5. **ASD evaluator**
   Scores out of 10; 8 or higher passes; best-scoring draft kept after 3 tries.

6. **SVG agent**
   Model draws SVG diagrams and NSL schematics, rasterized to PNG, with fallback vectors.

7. **Document exporter**
   Markdown builder, special outputs, quiz agent and accessibility tools assemble artifacts.

8. **Exports**
   PDF, PPTX, Braille, NSL diagrams and quizzes returned to the client.

QUALITY CONTROL FOR A SMALL LOCAL MODEL

The ASD rubric gate

Rule-based checks (score out of 10)

- Literal, factual sentences
- 2 to 3 short bullets per slide
- Uniform structure, no repetition
- No instructions or commands in slide text
- On topic, one example per slide
- Hindi vocabulary flagged in Nepali output

Pass mark

8 / 10

Below 8: the draft returns to the model with itemised corrections.

Up to 3 attempts. Best draft kept.

Heuristic gate, not a clinical measure.

EVERYTHING RUNS LOCALLY

Technology stack

Frontend	React Native (Expo); also served as a web app on port 8081
Backend	FastAPI served by Uvicorn
LLM engine	Local Ollama running qwen2.5:7b-instruct
Embeddings	nomic-embed-text via Ollama
Vector database	ChromaDB (persistent, local) over CDC guidelines
Visual assets	Model-drawn SVG (svg_agent.py) rendered by CairoSVG; no diffusion
Exporters	ReportLab (PDF), python-pptx (PPTX), Braille, NSL, quiz engine
Persistence	SQLAlchemy ORM with a local relational database
Packaging	Docker and Docker Compose

AI DOES REAL WORK IN THE CORE LOGIC

How it qualifies as "Build with AI"

Model output is parsed, validated, scored, stored and used to route control flow by program logic.

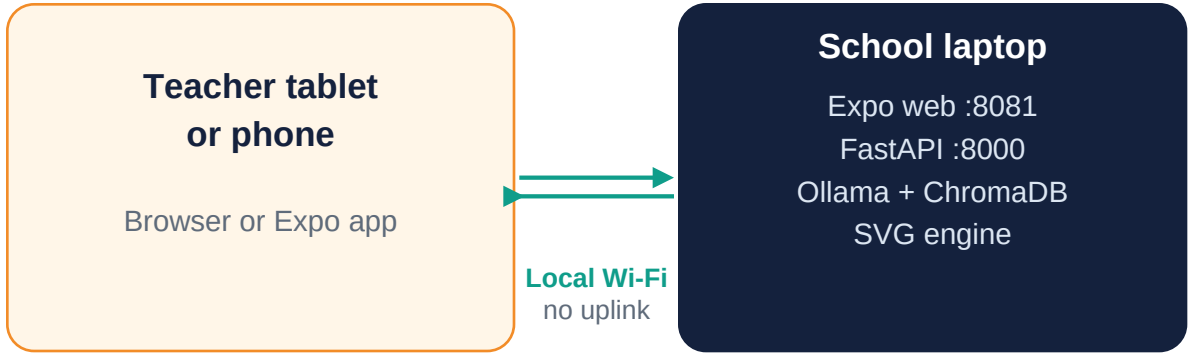
Where AI sits	How the system consumes it
Intent classification and extraction	JSON parsed; intent selects the branch; grade and subject become retrieval filters.
CoIEP writer-evaluator loop	Rule-based rubric scores each draft; below 8 triggers regeneration (max 3).
Embeddings and retrieval	ChromaDB similarity search with metadata filters; standards injected into prompts.
Curriculum extraction	Schema-constrained JSON validated with Pydantic, de-duplicated and stored.
Quiz and study sheets	Cleaned, de-duplicated, length-filtered with retries; rendered to PDF and Braille.
Vector illustration generation	SVG pattern-matched, rendered to PNG by CairoSVG, embedded in exports.

BUILT FOR CLASSROOMS WITHOUT INTERNET

Offline and air-gapped deployment

Staging (connected machine)

- Pull qwen2.5:7b-instruct and nomic-embed-text once.
- Build Docker images and export them with docker save.
- Vendor Python and frontend dependencies.
- Pre-ingest CDC guidelines into ChromaDB.
- Transfer by USB drive or local network.



Air-gap guarantees

- No outbound network calls at runtime.
- No external image-generation services.
- No telemetry or analytics SDKs.
- Student data stays in the local database and storage folder.

WHAT TEACHERS GET

Output formats

PDF

ReportLab + CairoSVG

Printable worksheets, IEP summaries, lesson plans

PPTX

python-pptx

Classroom slides with visual schedules

Braille

accessibility_tools.py

Embosser-ready or tactile-review transcriptions

NSL cards

accessibility_tools + svg_agent

Sign descriptions and schematic hand-sign illustrations

Quiz

quiz_agent.py

CDC-aligned assessments with answer keys

Illustrations

svg_agent + CairoSVG

Slide diagrams and sign schematics embedded in PDF and PPTX

All generated artifacts are written under backend/storage/.

HONEST SCOPE

Limitations and future work

Known limitations

- Teacher-assistance tool, not clinical or diagnostic. IEP goals are drafts for expert review.
- A 7B model can produce uneven output; the rubric is a heuristic.
- CPU-only generation can take tens of seconds to minutes; 8 to 16 GB RAM needed.
- Braille and NSL output must be reviewed by a Braille reader or signer.
- Single-school deployment; no multi-school sync.

Future improvements

- More Nepali vocabulary and CDC grade and subject coverage.
- Broader NSL vocabulary with animated previews.
- More disability profiles: visual, hearing, ADHD.
- Teacher feedback loop and a human-validated evaluation set.
- Local voice dictation and fully on-device tablet packaging.
- Encrypted storage, role-based access, offline multi-device sync.

Hours of preparation, reduced to minutes.

- Student data never leaves the school.
- Works for Braille and sign-language learners.
- No recurring cloud cost.

Built so every teacher, in every classroom, with or without internet, can support every learner.

github.com/UltimateDH/DraftBench